

is already known, after loading the calibration file.

Manual setting of the calibration factor sets the calibration file name to 'None Loaded' while the calibration factor is displayed in info box (3).

Measurement files

The files that are analysed by the system are measurement .wav files.

Load Meas .wav. This button loads the measurement .wav file for analysis. The measurement file must be a standard .wav format audio recording at sample rates of 22.05 kHz, 44.1 kHz, 48 kHz or 96 kHz.

Note: Only the first channel of a multichannel file is analysed and the rest are ignored.

Play Meas .wav (Windows only). This button will play the loaded measurement .wav file via Windows Media Player. This option is available in Windows versions only.

Speed. The speed buttons *Slow*, *Fast* and *Impulse* apply the ANSI, S1.4-1983 (R2006) standard exponential time-weighting functions for Lp plot analysis.

Note: Other plot analysis does not apply a time-weighting function.

Slow. This setting applies the 'Slow' time response with a time constant = 1.0 seconds. It is the most common setting used in standard sound-level meters, and is best used to estimate the average sound level of sounds with a limited dynamic range, such as speech.

Fast. This setting applies the 'Fast' time response with a time constant = 0.125 seconds. It is used for sound levels with a high dynamic range and fairly quick variations in the temporal response.

Impulse. This applies the 'Impulse' time response with a time constant = 0.035 seconds for rising signals and 1.5 seconds for falling signals. This function is used to accurately capture peak sound levels of impulsive noises such as gun shots.

Weighting

Weighting buttons A, C and Flat apply the ANSI S1.42-2001 standard

weighting or no weighting at all.

A. The A weighting filter is the most often used frequency response filter in acoustic analysis. For low-to-moderate sound levels (under about 80 dB), A-weighted sound levels are a better approximation to perceived loudness than any other weighted sound levels.

C. For higher sound levels (over about 90 dB), C-weighted sound levels are a better approximation to perceived loudness than A-weighted sound levels. If there is significant low-frequency content present in your acoustic signal, C weighting can be used in conjunction with A weighting.

Flat. For no frequency weighting (octave or 1/3-octave band analysis), use Flat weighting.

Analysis

There are three main analysis modes: time domain, frequency domain and mixed domain. Time domain analysis includes Lp mode and LEQ mode; frequency domain analysis includes Band mode and PSD mode; and mixed domain analysis is the Sgram or spectrogram mode.

Select the analysis mode by pressing the 'Analysis' button of any of the above modes and then press the yellow 'Analyse Data' button.

Note: If there is no measurement file loaded, a warning is displayed.

Lp mode. Select Lp mode to get the analysis of a standard non-integrating sound level meter.

Settings. 1. Select Lpplot→Set Plot Time Spacing from the main menu

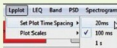


Fig. 2: Lpplot→Plot Scales

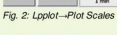


Fig. 3: LEQ→LEQ Integration Time

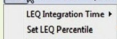


Fig. 4: LEQ→Noise Dose Criterion

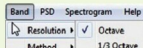


Fig. 5: Select Band→Resolution

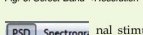


Fig. 6: PSD→FFT Size

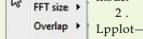


Fig. 7: Spectrogram→View

(1) [Fig. 2], to select the sample rate of the signal stimulated in Lp mode.

2. Select Lpplot→Plot Scales from the main menu (1) [Fig. 2], to select the plot scales as auto for auto scaling, or as manual for manually setting the X and Y minimum and maximum values.

LEQ mode. Se-

lect LEQ mode to get the analysis of a standard integrating sound level meter. LEQ takes a measurement file to compute a series of 100ms LEQs applying the selected frequency weighting.

Settings. 1. Select LEQ→LEQ Integration Time from the main menu (1) [Fig. 3], to set the length of the LEQ integrations, which are plotted and saved.

2. Select LEQ→Set LEQ Percentile from the main menu (1) [Fig. 3] to enter a level of percentile for computation and display.

3. Select LEQ→Plot Scales from the main menu (1) [Fig. 3], to select the plot scale as auto for auto scaling, or as manual for manually setting the X and Y minimum and maximum values.

4. Select LEQ→Noise Dose Criterion from the main menu (1) [Fig. 3], to open a dialogue box [Fig. 4] and enter the Exchange rate, Criterion and Threshold values.

Band mode. Select Band mode to compute the overall LEQ in the octave or 1/3 octave bands.

Settings. 1. Select Band→Resolution from the main menu (1) [Fig. 5], to select the resolution from octave or 1/3 octave.

2. Select Band→Method from the main menu (1) [Fig. 5], to select the method FFT or ANSI.

NC/RC mode. Select this mode to compute Noise Criterion (NC) and